

Validator-Guided Repair on a Shared Initial LLM Candidate: A Preregistered Paired Evaluation for Network Configuration Safety

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Abstract—We preregistered and executed a paired confirmatory experiment (PREREG-CAND-001-VER-GVR-2026-09-01) to measure whether permitting a deterministic network-configuration validator plus one automated repair changes the rate of validator-valid executable configurations relative to leaving the identical sampled LLM candidate unguarded (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` (hosted adapter `openai_responses`) and deterministic `netops_validator_v5` on $N = 200$ paired intents (`completed_episode_count = 400`; `model_calls_used = 200`), every shared initial candidate passed validator checks and no repairs were attempted or applied (`repair_attempt_count = 0`; `repair_applied_count = 0`). Observed outcomes: `baseline_success = 200/200`, `guarded_success = 200/200`; paired contingency $n11 = 200$, $n10 = 0$, $n01 = 0$, $n00 = 0$; exact McNemar two-sided $p = 1.0$; paired bootstrap percentile 95% CI (seed = 1729; 10,000 resamples) = $[0.0, 0.0]$. We present artifact-grounded methodology, execution accounting, representative validator diagnostics, compact contingency and repair summaries, and a focused discussion clarifying the causal limits of the executed estimand under `shared_initial_candidate` semantics and preregistered follow-on designs required to measure repair efficacy under independent sampling, higher difficulty, or adversarial inputs.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration sequences in network operations (NetOps). Erroneous sequences, syntactic mistakes, or fabricated fields can cause service disruption or policy violations when applied to devices. A common architectural mitigation is a generate→validate→repair (GVR) pipeline: generate candidate configuration(s), deterministically validate them against intent/topology/workflow constraints, and trigger automated repair when the validator finds faults. Prior work documents verifier-guided improvements in infrastructure-as-code and related code-generation domains and recommends grounding strategies to reduce hallucination in operational tasks [1–3]. However, whether verifier-guided pipelines reduce execution-level operational risk for network configurations remains an open empirical question because prior demonstrations often measure textual correctness or IaC test suites rather than executed device-state safety in packet-level or emulator contexts [3,4].

This paper reports a fully preregistered paired confirmatory experiment that isolates the downstream causal contribution of a deterministic validator and any permitted repair acting on the

exact same sampled LLM candidate (`shared_initial_candidate` semantics). Under these semantics baseline and guarded episodes for each intent begin from the identical initial generation; any paired difference can therefore only arise from deterministic validation and a permitted repair of that same candidate. Our primary research question is: under `shared_initial_candidate` semantics, does permitting a deterministic validator plus at most one automated repair change the proportion of validator-valid executable configurations relative to leaving the same initial candidate unguarded? The contribution is an artifact-grounded report of the executed estimand with complete execution accounting, exact inferential statistics, representative validator diagnostics, and precise recommendations for preregistered follow-on designs that can measure repair efficacy when independent sampling, elevated difficulty, or adversarial inputs are employed.

II. RELATED WORK

Hallucination and factual unreliability in LLM outputs are well documented and motivate grounding, retrieval, and verification strategies for operational tasks [1]. Several recent surveys synthesize these failure modes and recommend evidence-grounding or verification as primary mitigations; they emphasize that hallucination manifests across domains and that mitigation effectiveness depends strongly on task framing and the grounding mechanism used [1].

Retrieval-augmented and evidence-grounded pipelines have been shown to reduce hallucination in operations-style question answering and documentation tasks, primarily by constraining the model with external factual context and by post-generation grounding checks [2]. These evaluations typically measure textual fidelity (e.g., citation correctness, answer grounding) rather than executable device-state safety. As a result, reductions in text-level hallucination do not directly imply reductions in execution risk when generated artifacts are translated into device CLI and applied to network state.

Generate→validate→repair (GVR) architectures have been proposed and demonstrated in adjacent domains such as infrastructure-as-code (IaC) and program synthesis, where deterministic or policy-guided verifiers detect rule violations and either supply feedback to the generator or guide repair loops [3]. TerraFormer and related verifier-guided IaC systems operationalize a feedback loop that fine-tunes or constrains generation via verifier signals; reported gains are measured against

IaC test suites and policy checks rather than execution on heterogeneous device families. Key methodological elements highlighted by these works include (a) verifier expressivity (what constraints and dependency rules are implemented), (b) repair strategy (single corrective edit, multi-step rewriting, or fine-tuning), and (c) how verifier feedback is integrated (in-prompt, tool call, or offline retraining) [3].

NetOps-specific evaluations of LLM capability concentrate on intent translation accuracy, syntax compliance, and component-level parsing rather than end-to-end execution safety in simulators or testbeds [4]. These studies typically report metrics such as command-level BLEU/similarities, slot accuracy, or human adjudication of intent fidelity; they provide useful capability baselines but leave open how textual errors map to executable misconfigurations or service disruption when applied to device state. AIOps surveys further underscore the gap between capability benchmarking and execution-aware evaluation: they call for standardized execution-level benchmarks and validator fidelity measurements that tie generator outputs to concrete operational safety outcomes [5].

From a methodological standpoint, prior literature therefore separates three evaluation axes: (1) generator sampling variability (how often first-pass candidates are correct), (2) validator fidelity (whether the verifier captures the operational invariants and dependency rules relevant to execution), and (3) repair efficacy (whether permitted repairs transform invalid candidates into valid, executable ones). Many published demonstrations conflate (1) and (3) by sampling independently across conditions or by evaluating repaired outputs only at the text level. The paired, shared_initial_candidate semantics used in our preregistration explicitly isolate axis (2) and (3) acting on a single sampled candidate: baseline and guarded arms begin from the identical generation so any difference must come from deterministic validation and permitted repair rather than from independent sampling.

This distinction clarifies why IaC results are informative but not decisive for NetOps execution safety. TerraFormer-style verifier-guided improvements (IaC) indicate that verifier feedback can materially improve generator correctness in structured code domains [3]. However, network device configuration introduces additional complexity: vendor CLI differences, runtime protocol interactions, timing-sensitive sequences, and multi-device workflows. Existing NetOps capability studies and AIOps surveys document translation ability and recommend validator uplift but, crucially, do not provide standardized execution-level outcomes that map generator errors to runnable device states—creating the empirical gap that motivated our execution-focused preregistered study [4,5].

Finally, the literature highlights practical evaluation considerations that our experiment operationalizes: use deterministic task manifests and contamination checks to avoid corpus leakage, record provider audit traces and model configuration details (including explicitly noting an unset temperature rather than assuming deterministic sampling), and archive per-episode validator traces (validation_before/validation_after, parsed/required command counts,

repair_applied flags) to permit exact replay and independent verification [2–5]. These provenance and scoring practices are commonly recommended but rarely presented end-to-end in a preregistered execution bundle that also enforces a narrowly scoped estimand—our study adopts these artifact practices to make the estimand and its limits explicit and reproducible.

III. METHODOLOGY

Preregistration and primary estimand. The study preregistration (PREREG-CAND-001-VER-GVR-2026-09-01) froze the primary estimand and analysis before any model calls. Primary estimand: `paired_success_rate_difference_guarded_minus_baseline` (per-intent paired difference in proportion of validator-valid executable configurations) under `shared_initial_candidate` semantics. Design: deterministic paired sample of `task_count = 200` intents producing `planned_episode_count = 400` episodes. Generation semantics: exactly one sampled initial generation per pair (shared) and deterministic reuse by baseline and guarded episodes. Guarded episodes allowed up to one automated repair (`maximum_repair_calls_per_task = 1`). Primary test: exact two-sided McNemar test; paired bootstrap (seed = 1729; 10,000 resamples) for a percentile 95% CI.

Task bank and contamination controls. A deterministic task generator produced the manifest of 200 synthetic `intent_configuration_repair_v1` tasks. Each manifest entry records `initial_state`, `required_sequence`, difficulty annotations, and a `generation_seed` for deterministic replay. Preexecution contamination checks (SHA-256 exact-match and n-gram similarity thresholds) were applied; no exact-match contamination exclusions were flagged and the confirmatory sample executed intact. Contamination and task manifest artifacts are archived in the execution evidence bundle.

Model calls, generation semantics, and provider audit. Execution used the declared generator labeled `gpt-5-mini` via the hosted adapter `openai_responses` and consumed `model_calls_used = 200` (one shared initial generation per pair). The archived `model_configuration.json` records `declared_model_version = "gpt-5-mini"` and `temperature = null/unset`; we explicitly state that `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`. Provider audit traces record `hosted_model_call_event_count = 200`, `cache_miss_count = 200`, and `stage_counts` showing `shared_initial_generation = 200` consistent with the preregistered `shared_initial_candidate` semantics.

Deterministic validator and repair policy. Scoring used `deterministic_netops_validator_v5` with `scoring_rule = "full_intent_constraint_validation."` The validator parses candidate commands, applies task `workflow_policies` and `required_sequence` checks, and produces structured validator traces with `validation_before` and `validation_after` subtables, `parsed_command_count`, `required_command_count`, `normalized_configuration`, `violation_codes`, and a `repair_applied` flag. For guarded episodes one automated repair attempt was permitted; the repair, if invoked, would produce

a new candidate which the validator would re-validate. Under `shared_initial_candidate` semantics only the guarded arm could alter the identical initial candidate via the permitted repair; the baseline arm reused the same candidate unchanged.

Mapping to outcomes and missingness. `Validator.valid = true` mapped to a binary "executable/valid" outcome for each episode. Primary analysis adhered to the preregistered `complete_pair_primary` policy (exclude incomplete pairs). A sensitivity analysis (failure-as-zero) was preregistered to treat missing or failed episodes as unsafe. Multiplicity: single confirmatory McNemar test; exploratory subgroup analyses (e.g., by difficulty patterns) were labeled exploratory and corrective procedures (Benjamini–Hochberg) were specified if applied.

IV. RESULTS

Execution accounting and deterministic reconciliation. The execution completed exactly as preregistered: `planned_episode_count = 400`, `completed_episode_count = 400`, `failed_episode_count = 0`, and `model_calls_used = 200` (one shared initial generation per pair). Analysis `missingness_summary` reports `complete_pairs = 200` with zero failed or unscorable episodes. Deterministic reconciliation records `hosted_model_call_event_count = 200`, `completed_hosted_model_call_event_count = 200`, `cache_miss_count = 200`, and `stage_counts = {"shared_initial_generation": 200}`; `cross_condition_cache_key_reuse_observed = false` and all deterministic consistency checks passed.

Primary binary outcomes and contingency. The validator mapped each episode to a binary executable outcome (`validator.valid`). Condition summaries show `baseline_successes = 200/200` and `guarded_successes = 200/200` (`success_rate = 1.0` for both arms). The paired 2×2 contingency table is `n11 = 200`, `n10 = 0`, `n01 = 0`, `n00 = 0`; exact McNemar two-sided $p = 1.0$. The paired bootstrap (seed = 1729; 10,000 resamples) produced a percentile 95% CI for the paired difference of [0.0, 0.0]. These files are archived (`analysis/condition_summary.csv` and `analysis/paired_contingency_table.csv`) and exactly reproduce the contingency and CI numbers reported here.

Repair activity, validator diagnostics, and representative examples. Across the confirmatory sample the validator flagged no violations on any shared initial candidate; as a result `repair_attempt_count = 0` and `repair_applied_count = 0`. Representative per-episode scoring JSON artifacts illustrate the pattern. For example, task-000004 (pair_id pair-000004) `shared_initial_candidate` and both condition responses were identical:

```
interface edge2 admin down interface edge2 vlan 40 interface edge2 admin up interface edge1 admin down
```

For task-000004 the archived validator trace shows `parsed_command_count = 4`, `required_command_count = 4`, `observed_touched_field_count = 3`, and `validation_after.valid = true`. Similarly, task-000005 shows `parsed_command_count = 3` and `required_command_count = 3` with `validation_after.valid = true`; task-000006 shows `parsed_command_count = 6` and `required_command_count =`

6 with `validation_after.valid = true`. These per-episode diagnostics (`execution/scoring/task-00000X-*.json`) demonstrate that the validator parsed the expected command set, matched required sequence checks, and reported no violation codes for the sampled candidates. Because the validator never produced a violation, the guarded repair path was never invoked and could not alter any outcome.

Contamination and provider audit summary. Preexecution contamination checks found no exact-match contamination exclusions; `analysis/contamination_summary.csv` is empty. Provider audit and execution manifests corroborate the sampling semantics: `hosted_model_call_event_count = 200`, `cache_miss_count = 200`, and `stage_counts` indicating exactly one `shared_initial_generation` per pair. No infrastructure retries or failures occurred; `analysis/analysis_log.jsonl` documents the paired analysis completion and bootstrap parameters (`bootstrap_resamples = 10000`; `bootstrap_seed = 1729`).

Compact repair summary (explicit). In concise preregistered terms: `total_pairs = 200`; `repair_attempt_count = 0`; `repair_applied_count = 0`. The empty repair counts fully explain the degenerate paired contingency (all pairs valid before any repair opportunity). All artifacts supporting these counts are archived in the evidence bundle (notably the `execution/scoring` JSONs, `execution/task_manifest.jsonl`, `analysis/paired_contingency_table.csv`, and `analysis/condition_summary.csv`) and permit independent verification that no repairs were produced or applied in guarded episodes.

V. DISCUSSION

Interpreting the executed estimand. The `shared_initial_candidate` semantics were intentionally chosen to isolate the causal effect of deterministic validation and any permitted repair acting on the exact same generator output. The executed estimand therefore answers a narrowly defined causal question: when baseline and guarded episodes begin from the identical sampled candidate, can deterministic validation and at most one automated repair change the `validator-valid/executable` outcome? The observed null effect (paired difference = 0.0) shows that, for this task bank and this single sampled candidate per pair, the validator found no violations and the permitted repair path was never exercised. Importantly, this does not evaluate whether guarded pipelines reduce first-pass generator error rates under independent sampling or different model temperatures; such claims would require a different preregistered estimand and execution.

Concrete validator behavior and why the guarded path was not invoked. The `deterministic_netops_validator_v5` used in scoring enforces a structured sequence of checks: lexical/command parsing, mapping commands to `normalized_configuration`, counting `parsed_command_count` against `required_command_count` from the task manifest, and evaluating workflow policies such as "minimum_active_in_groups" and "must_be_down_for_fields." The archived per-episode JSON traces show `parsed_command_count == required_command_count`

and `validation_after.valid == true` for representative tasks (e.g., task-000004: `parsed_command_count = 4`, `required_command_count = 4`), which means the validator accepted the shared initial candidate as satisfying the syntactic and workflow constraints recorded in the task manifest. Because the validator reported zero `violation_codes` across the confirmatory sample, the guarded branch's single permitted automated repair had no trigger and therefore `repair_attempt_count = 0` and `repair_applied_count = 0`. This artifact-level diagnostic fully explains the degenerate paired contingency ($n_{11} = 200$) observed in the results.

Statistical and power implications given zero discordance. The exact McNemar two-sided $p = 1.0$ and bootstrap CI = $[0.0, 0.0]$ are deterministic consequences of $n_{10} = n_{01} = 0$. From an inferential perspective, zero discordant pairs carry no information about the sign or magnitude of a potential repair benefit under sampling regimes that would produce validator failures. The preregistered power analysis assumed nonzero baseline unsafe rates (sensitivity checks considered baseline unsafe rates in $\{0.2, 0.3, 0.4\}$ and target effects $\approx 0.08\text{--}0.12$). Those assumptions remain necessary: to detect an absolute risk reduction on the order of 10 percentage points with $\sim 80\%$ power requires an observable rate of validator failures under the baseline sampling distribution. In practice this means either (a) selecting task items with a higher expected difficulty/failure label, (b) altering generation sampling to increase first-pass diversity (e.g., nonzero temperature), or (c) injecting adversarial perturbations so that the guarded repair path is exercised. Without such adjustments the paired shared_initial_candidate design can produce informative nulls when the validator accepts nearly all sampled candidates, as occurred here.

Construct validity of the validator abstraction. Deterministic_netops_validator_v5 provides strong syntactic and workflow-policy checks and records structured diagnostics that make its operation auditable in artifact traces. However, construct validity limits remain: the validator maps candidate commands to an asserted final_state and enforces required_sequence and policy constraints, but it does not execute vendor CLI on hardware nor perform packet-level emulation of run-time protocol interactions. As a result, certain dynamic failure modes (e.g., transient routing convergence, BGP session flaps, or timing-dependent failover behavior) are outside the validator's scope. The evidence bundle therefore supports a qualified conclusion only about validator-level syntactic/workflow correctness for the sampled candidates, not about runtime-observable failures that a live device or packet-level emulator might reveal. Representative task annotations in the manifest (difficulty levels labeled "very_hard" and an isolated "extreme" instance) indicate the generator was challenged by multi-step required_sequences, yet the validator still accepted the single sampled generations; this suggests that either the generator produced compliant plans for these synthetic tasks or that the task bank difficulty distribution did not produce the class of subtle errors that would need repair.

Operational implications and conservative interpretation.

Practitioners evaluating GVR pipelines should note two operational lessons from these artifacts. First, an observed `guarded==baseline null` under shared_initial_candidate semantics does not imply that a GVR pipeline would be unhelpful under standard deployment conditions (where multiple independent samples, different temperatures, or diverse models are used). Second, artifact diagnostics can reveal the immediate cause of a null: here the cause was absence of validator detections (repair path never invoked). Operational evaluation should therefore pair estimator choices with task selection that yields nontrivial validator-failure rates when the goal is to measure repair efficacy. For safety-critical deployments, a practical path is to strengthen validator fidelity (e.g., integrate vendor CLI semantics, run candidate changes against a packet-level emulator or staged lab, or add dynamic checks for protocol convergence) so that the validator better approximates the operational harm surface that repairs must address.

Methodological recommendations grounded in archived evidence. The execution and manifest artifacts identify specific, artifact-traceable modifications for follow-on preregistered experiments that would place the repair mechanism under test: (1) relax shared_initial_candidate semantics to generate independent first-pass candidates per condition so that generator variability can produce baseline failures that repairs might fix (this requires changing generation_semantics and re-defining the estimand); (2) increase the expected baseline failure rate via task selection (prefer manifest entries annotated "extreme" or compose adversarial task variants) so that repairs are triggered with nontrivial probability; (3) preregister explicit sampling parameters (temperature sweeps) instead of leaving temperature null/unset; the archived model_configuration.json makes clear that temperature = null in this run and that null/unset does not imply deterministic sampling; (4) augment the validator by integrating higher-fidelity execution backends (vendor CLI emulation or packet-level simulation) to raise construct validity for runtime failure modes; and (5) increase permitted repair iterations or diversify repair strategies so repairs have more opportunity to converge on a valid candidate when initial violations are detected. Each of these recommendations follows directly from the execution artifacts (e.g., stage_counts showing only shared_initial_generation = 200 and the repair_applied = false flags across the sample) and the preregistration sensitivity analysis that assumed nonzero baseline unsafe rates.

Reproducibility and artifact auditing. The artifact bundle contains per-episode validator traces (validation_before/validation_after), raw shared initial responses, the execution manifest, model_configuration.json (declared_model_version = "gpt-5-mini", temperature = null), and analysis/condition_summary.csv and analysis/paired_contingency_table.csv. These artifacts permit independent auditors to verify that: (a) exactly one shared initial generation per pair was produced and reused in both arms (provider audit: shared_initial_generation = 200); (b) validator.valid was true for all episodes and no repair prompts or repair traces were created (repair_attempt_count = 0);

and (c) the exact contingency counts driving the McNemar test are reproducible. Importantly, auditors should not infer deterministic model sampling from temperature = null; the execution manifest explicitly records that temperature was unset and that sampling nondeterminism cannot be excluded for the single sampled generation per pair.

Threats to external validity and next steps. The principal external validity limits are single-model scope (declared_model_version = "gpt-5-mini" via the archived adapter), synthetic deterministic task bank, and the validator abstraction described above. These limits are documented in the registered limitations and in the evidence bundle. The most direct path to broader external validity is a controlled factorial follow-on that crosses (i) independent vs shared generation semantics, (ii) model temperature and model family, and (iii) validator fidelity (syntactic/workflow vs emulator-integrated). Such a design would separate generator error rates from repair efficacy and allow estimation of the number-needed-to-repair under realistic sampling variability. All of these modifications are prespecified in the paper's preregistered follow-on recommendations and can be implemented while preserving the same deterministic task manifest structure for continuity of comparison.

Summary. The expanded artifact diagnostics show that the run produced an informative but narrowly scoped result: deterministic validation found no violations for the single sampled candidate per intent, so the permitted guarded repair path was never exercised and could not change outcomes. This clarifies the causal limit of the executed estimand and points directly to preregistered experimental modifications needed to measure repair efficacy under conditions that produce validator failures and thus allow repairs to act.

VI. LIMITATIONS

- Shared_initial_candidate semantics: the design produced exactly one sampled initial generation per intent and reused it across baseline and guarded conditions; this measures validator/repair effects only and cannot detect differences caused by independent per-condition sampling or prompt engineering.
- Temperature unset (temperature = null) in the archived model configuration: null/unset is not evidence of deterministic sampling and does not imply temperature = 0; sampling nondeterminism may still have occurred during the single sampled generation.
- Single model and single hosted provider: results reflect gpt-5-mini under the archived adapter; generalization to other models or model families is unsupported by these artifacts.
- Synthetic task bank and validator abstraction: tasks were synthetic 'intent_configuration_repair_v1' items and the deterministic validator is an abstraction; realistic vendor CLI idiosyncrasies, emergent runtime interactions, and hardware effects were not executed on live devices.
- No observed validator failures: all initial candidates were validator-valid, so the experiment could not assess the

repair step's effectiveness; the null effect therefore reflects the task bank and sampling semantics rather than evidence that GVR is unnecessary in general.

VII. CONCLUSION

We executed a fully preregistered paired confirmatory experiment that measures the downstream effect of a deterministic validator plus an allowed single automated repair on the exact same sampled LLM candidate per intent (shared_initial_candidate semantics). Using the declared generator gpt-5-mini and deterministic_netops_validator_v5 on 200 paired intents, every shared initial candidate passed validation and no repairs were attempted or applied; guarded and baseline outcomes were identical for the executed estimand (paired difference = 0.0; McNemar $p = 1.0$; paired bootstrap CI = [0.0, 0.0]). This artifact-grounded null result demonstrates an estimand boundary: under shared_initial_candidate semantics and the executed task bank the guarded pipeline could not be exercised and therefore could not change outcomes. It does not imply that GVR pipelines are unnecessary generally. Preregistered follow-on experiments that (1) remove shared_initial_candidate semantics, (2) increase task difficulty or inject adversarial inputs, (3) set explicit sampling parameters, and (4) raise validator fidelity are required to empirically evaluate repair efficacy under realistic sampling variability and adversarial conditions. The archived artifacts and reconciliation files enable exact reproduction of the executed estimand and provide a secure base for precise preregistered extensions.

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One-line reiteration of the primary estimand limit: the preregistered primary estimand intentionally used shared_initial_candidate semantics; evaluating per-condition independent sampling requires a different preregistration and analysis plan (explicitly recommended above).

DISCLOSURE STATEMENT

Disclosure (concise): The human Research Director selected the broad topic family (Generative AI / LLMs for NetOps) before the autonomy lock. After the autonomy lock the autonomous system performed literature review, experiment selection, preregistration (PREREG-CAND-001-VER-GVR-2026-09-01), code generation, execution, deterministic validation, automated analysis, and manuscript drafting without manual human

editing of technical content. Declared model used in execution: gpt-5-mini via hosted adapter 'openai_responses' (execution used model_calls_used = 200; archived model_configuration.json records temperature = null/unset; null/unset is not evidence of deterministic sampling and does not imply temperature = 0). Generation semantics executed: shared_initial_candidate — exactly one sampled initial generation per intent was produced and deterministically reused for both baseline and guarded conditions. Validator: deterministic_netops_validator_v5 scored candidates using 'full_intent_constraint_validation' and a single automated repair iteration was permitted in guarded episodes; in this run the validator flagged no violations and no repairs were applied (repair_attempt_count = 0; repair_applied_count = 0). Execution accounting: completed_episode_count = 400 (200 paired intents) completed without preregistration deviations. Master prompt immutable reference: provenance/master_prompt.txt — SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba. Pre-lock infrastructure development and rehearsal occurred but pre-lock artifacts were not used as empirical evidence in this final run. After the autonomy lock no human manually rewrote technical results, manuscript text, figures, or references.

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